

Heart Rate Detection from Ballistocardiogram Signals: A Julia Implementation



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Outline



- Introduction
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- Implementation
- Evaluation Metrics

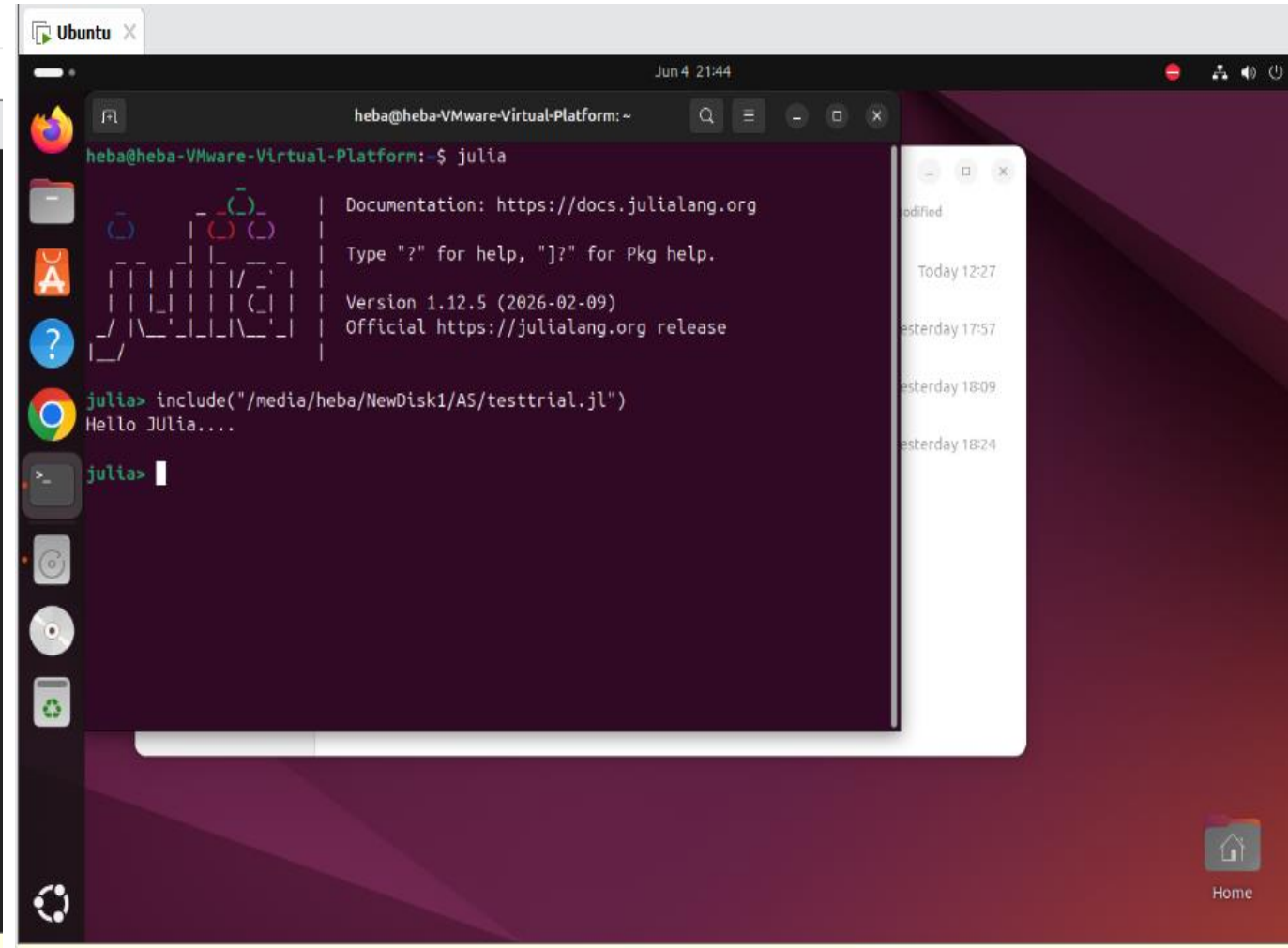
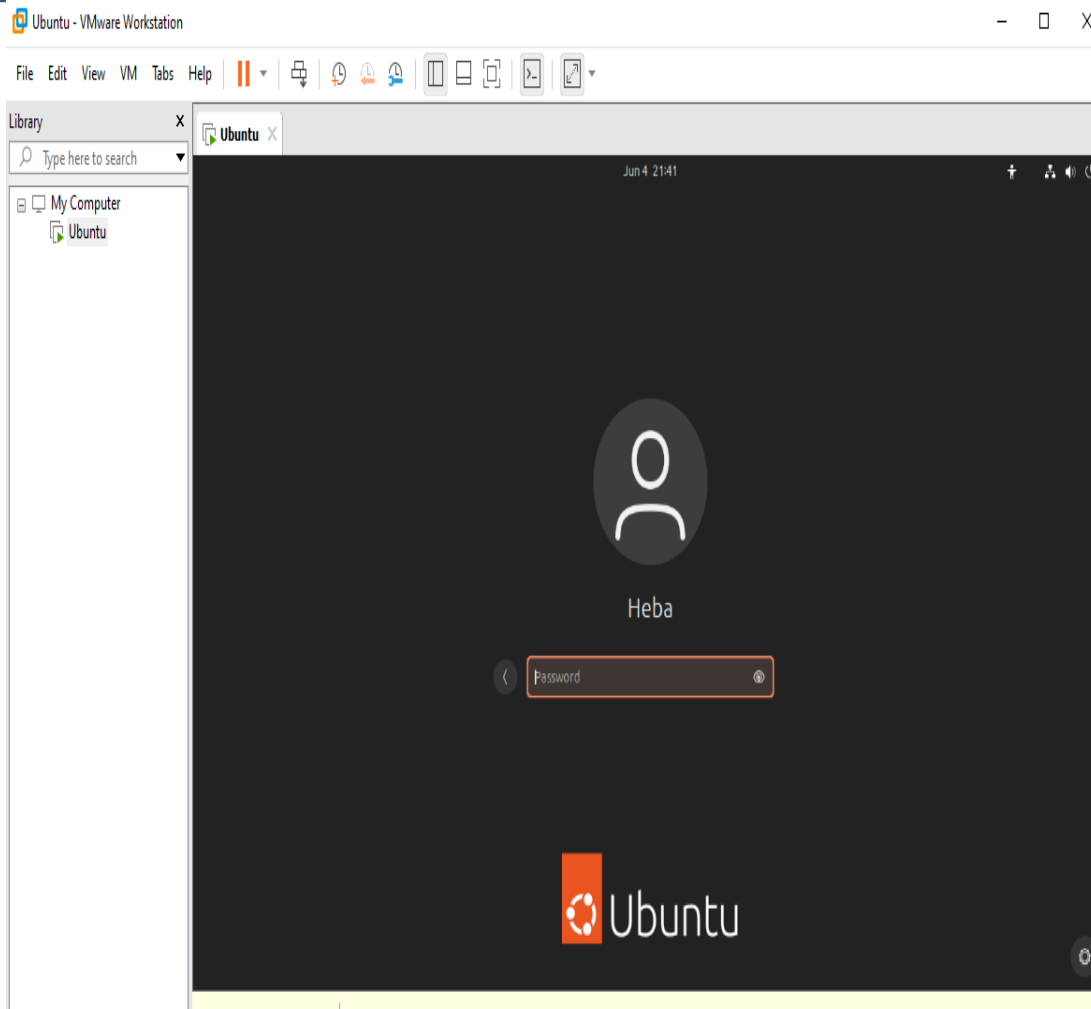
Introduction

- Estimate Heart Rate (HR) from Ballistocardiography (BCG) signals, then statistically evaluate the estimation accuracy against reference ECG recordings across multiple patients.
- This project is implemented in **Julia**, based on an existing reference pipeline originally written in Python and published on Code Ocean.

Environment Setup

- Installed VMware Workstation Pro to create a virtual machine environment.
- Mounted and configured Ubuntu Linux on VMware.
- Installed Julia on Ubuntu.
- Used the Julia REPL (Read–Eval–Print Loop) to run and test Julia scripts.

Environment Setup



Implementation (Phase1)

1. Convert reference code (from Code Ocean) to Julia
 - └─ Using single BCG sample data file ($fs = 50$ Hz)
2. Modular code design
 - └─ Each processing stage implemented as an independent, reusable function
 - └─ Entry point : "mainEntry.jl"
 - └─ Signal processing : "data_preprocessing.jl"
 - └─ Evaluation : "metrics_computation.jl",
"p_value_calculations.jl",
"pearson_relation_coeff.jl",
"regression_plot_and_bland_altman_plot.jl"

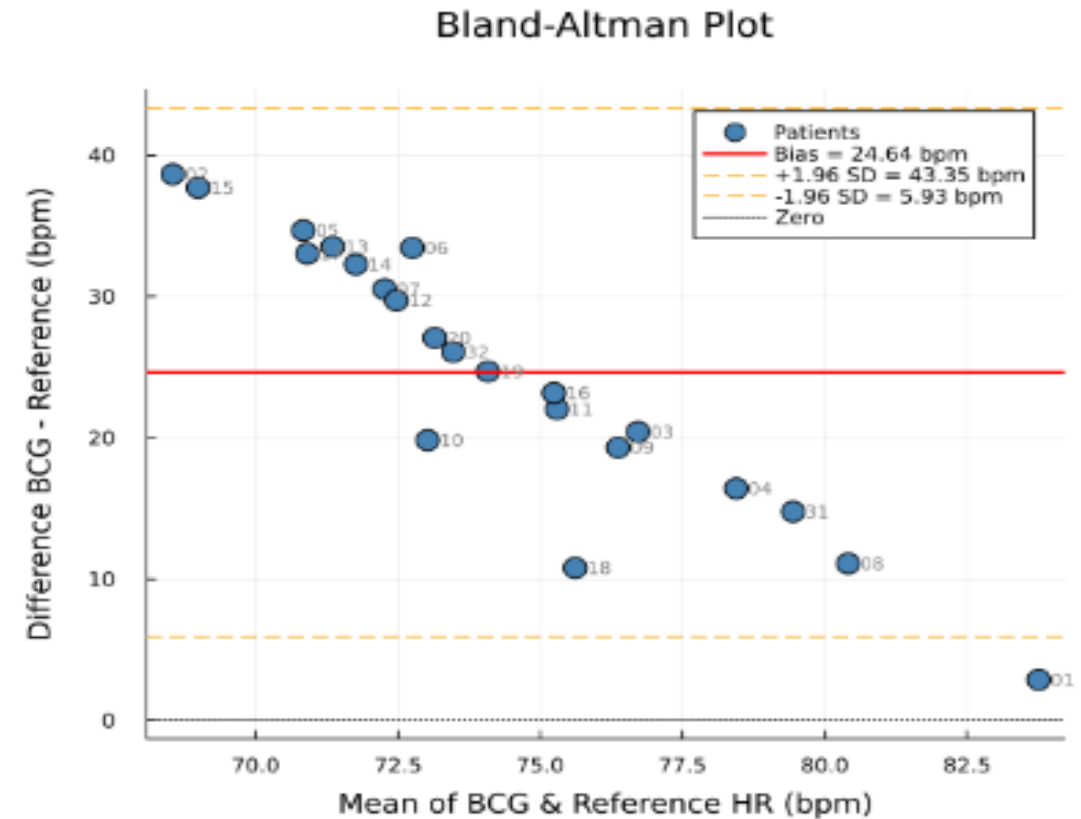
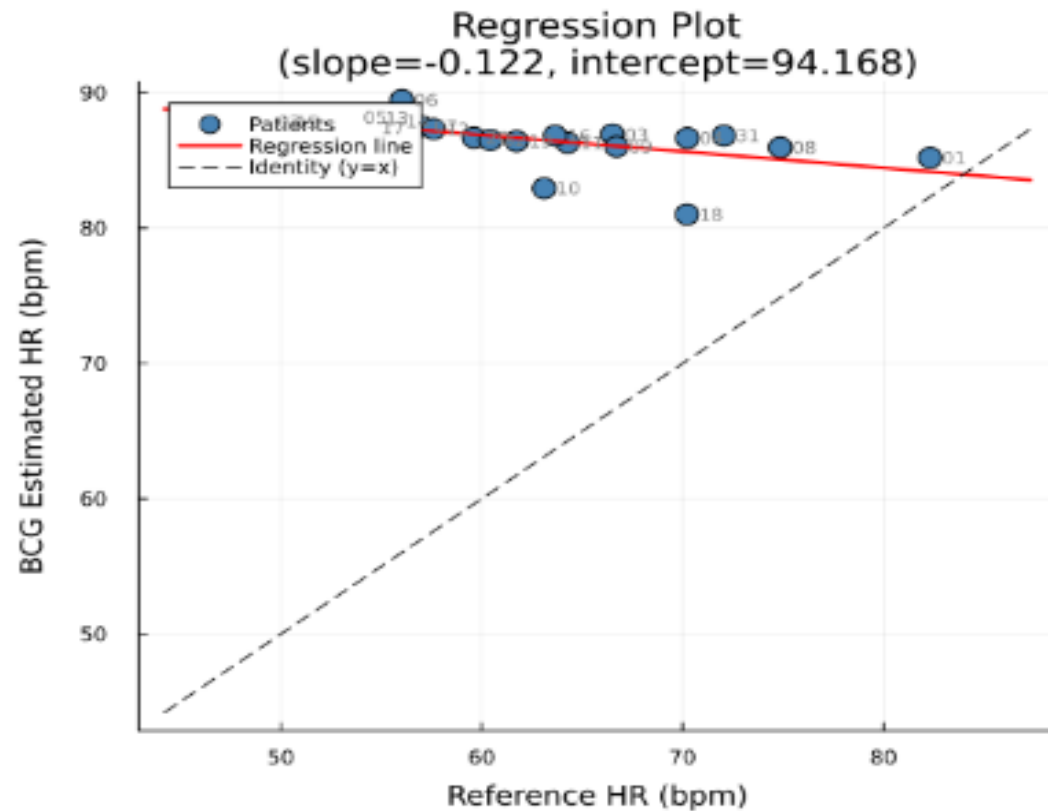
Implementation (Phase 2)

1. Run the obtained Julia code on a single file from the new dataset
 - └─ Single file validation (fs = 140 Hz)
2. Run the code across all patients from the new dataset
3. Calculate metrics by comparing BCG-estimated HR against reference ECG signals
 - └─ MAE, RMSE, MAPE, Pearson r, p-value, Regression Plot, Bland-Altman Plot

Evaluation Metrics

Metric	Value	Metric	Value
MAE	24.642 bpm	Pearson r	-0.5694
RMSE	26.348 bpm	t-stat	-3.0977
MAPE	42.339 %	p-value	0.002837

Regression plot and Bland–Altman plot





Questions

Thank You!